WORKED SOLUTION

STEP II 2020, Question 3

Pure • Log-Concave Sequences

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Property L says that each ratio u_{r+1}/u_r is at most the one before it, so once the sequence stops rising it can never start rising again. That is exactly unimodality. In part (ii) the recurrence has a repeated root α in its auxiliary equation, and the combination $u_r - \alpha u_{r-1}$ turns it into a simple geometric sequence. Everything else follows from that one substitution.

Solution

Part (i)

Suppose the sequence has property L and $u_{r-1} \geq u_r$, where $2 \leq r \leq n-1$. Since $u_{r-1} > 0$, we can divide property L by it:

$$u_{r+1} \leq \frac{u_r^2}{u_{r-1}} = u_r \cdot \frac{u_r}{u_{r-1}} \leq u_r,$$

where the last step uses $u_r/u_{r-1} \leq 1$ and $u_r > 0$. So $u_{r-1} \geq u_r \implies u_r \geq u_{r+1}$.

Now take any sequence with property L. If $u_1 \leq u_2 \leq \cdots \leq u_n$, it is unimodal with $k = n$. Otherwise there is a first index k with $u_k > u_{k+1}$ (so $k \leq n-1$). By the choice of k,

$$u_1 \leq u_2 \leq \cdots \leq u_k.$$

Since $u_k \geq u_{k+1}$, the implication just proved (with $r = k+1$) gives $u_{k+1} \geq u_{k+2}$; applying it again with $r = k+2$ gives $u_{k+2} \geq u_{k+3}$, and so on by induction up to $u_{n-1} \geq u_n$. Hence $u_k \geq u_{k+1} \geq \cdots \geq u_n$, and the sequence is unimodal.

Part (ii)

Write $v_r = u_r - \alpha u_{r-1}$ for $2 \leq r \leq n$. For $3 \leq r \leq n$, the recurrence gives

$$v_r = 2\alpha u_{r-1} - \alpha^2 u_{r-2} - \alpha u_{r-1} = \alpha\left(u_{r-1} - \alpha u_{r-2}\right) = \alpha v_{r-1}.$$

So (v_r) is geometric with ratio α, and $v_r = \alpha^{r-2}v_2$, that is,

$$u_r - \alpha u_{r-1} = \alpha^{r-2}(u_2 - \alpha u_1) \qquad (2 \le r \le n).$$

(For $r = 2$ this is trivially true, and each step up in r multiplies both sides by α.)

For $2 \le r \le n-1$ we have $3 \le r+1 \le n$, so we may substitute $u_{r+1} = 2\alpha u_r - \alpha^2 u_{r-1}$:

$$u_r^2 - u_{r-1}u_{r+1} = u_r^2 - 2\alpha u_r u_{r-1} + \alpha^2 u_{r-1}^2 = (u_r - \alpha u_{r-1})^2.$$

Positive and unimodal. Suppose $u_2 > \alpha u_1 > 0$. Since $\alpha > 0$, this gives $u_1 > 0$, and $u_2 > 0$ too. For the later terms, the first identity gives

$$u_r = \alpha u_{r-1} + \alpha^{r-2}(u_2 - \alpha u_1) > \alpha u_{r-1},$$

because $\alpha^{r-2} > 0$ and $u_2 - \alpha u_1 > 0$. So if $u_{r-1} > 0$ then $u_r > 0$, and by induction every term is positive. The second identity now gives $u_{r-1}u_{r+1} = u_r^2 - (u_r - \alpha u_{r-1})^2 \le u_r^2$ for $2 \le r \le n-1$. That is property L for a sequence of positive terms, so the sequence is unimodal by part (i).

The case $u_1 = 1$, $u_2 = 2$. Let $w_r = (2-r)\alpha^{r-1} + 2(r-1)\alpha^{r-2}$. Then $w_1 = 1 = u_1$ and $w_2 = 2 = u_2$. With these starting values the first identity reads

$$u_r = \alpha u_{r-1} + (2-\alpha)\alpha^{r-2}.$$

Suppose $u_{r-1} = w_{r-1}$ for some r with $3 \le r \le n$. Then

$$\begin{aligned} u_r &= \alpha\left[(3-r)\alpha^{r-2} + 2(r-2)\alpha^{r-3}\right] + 2\alpha^{r-2} - \alpha^{r-1} \\ &= (2-r)\alpha^{r-1} + 2(r-1)\alpha^{r-2} = w_r, \end{aligned}$$

so by induction $u_r = (2-r)\alpha^{r-1} + 2(r-1)\alpha^{r-2}$ for $1 \le r \le n$. (We could equally run a two-step induction using the original recurrence; the first-order form is quicker.)

The maximum when $\alpha = 1 - \frac{1}{N}$. Here $\frac{1}{2} \le \alpha < 1$, so $u_2 = 2 > \alpha = \alpha u_1 > 0$, and by the work above the sequence is positive and unimodal. To find the peak, look at consecutive differences. From $u_{r+1} = \alpha u_r + (2-\alpha)\alpha^{r-1}$ and $\alpha - 1 = -\frac{1}{N}$,

$$u_{r+1} - u_r = (2-\alpha)\alpha^{r-1} - \frac{u_r}{N} = \frac{\alpha^{r-2}}{N}\Big[N\alpha(2-\alpha) - (2-r)\alpha - 2(r-1)\Big].$$

Now $N\alpha = N - 1$ and $2 - \alpha = \frac{N+1}{N}$, so $N\alpha(2-\alpha) = N - \frac{1}{N}$, and $(2-r)\alpha = 2 - r - \frac{2-r}{N}$. The bracket simplifies to

$$N - \frac{1}{N} - 2 + r + \frac{2-r}{N} - 2r + 2 = (N-r) - \frac{r-1}{N} = \frac{N^2 + 1 - r(N+1)}{N},$$

so

$$u_{r+1} - u_r = \frac{\alpha^{r-2}}{N^2}\Big(N^2 + 1 - r(N+1)\Big).$$

Ask yourself Where exactly does $N^2+1-r(N+1)$ change sign, and could it ever be zero?

It is zero only at $r = \dfrac{N^2+1}{N+1} = N-1+\dfrac{2}{N+1}$. Since $N \geq 2$, we have $0 < \frac{2}{N+1} < 1$, so this value lies strictly between the integers $N-1$ and N. As $\alpha^{r-2} > 0$:

- for $r \leq N-1$, $u_{r+1} > u_r$;
- for $r \geq N$, $u_{r+1} < u_r$.

Hence $u_1 < u_2 < \cdots < u_N$ and $u_N > u_{N+1} > \cdots > u_n$. Since $N \leq n$, the term u_N exists, and u_r is largest when $r = N$ (strictly, with no ties).

Checks

Take $N = 5$, so $\alpha = \frac{4}{5}$. The recurrence gives 1, 2, 2.56, 2.816, 2.8672, 2.78528, 2.62144, …, rising to its peak at $r = 5$ and falling afterwards, as claimed. The difference formula predicts $u_6 - u_5 = \alpha^3(26-30)/25 = -0.08192$, which matches. Also $u_3 = 2\alpha \cdot 2 - \alpha^2 = 2.56$ agrees with the closed form $-\alpha^2 + 4\alpha$. The identities and the location of the peak were also confirmed by computer, in exact arithmetic, for many values of α, u_1, u_2 and for N up to 29.

Common slips

- In part (i), dividing by u_{r-1} without noting that it is positive. The argument fails for sequences with negative terms.
- Forgetting to show the terms are positive in part (ii) before appealing to part (i), which needs positivity.
- Showing only that $u_{N+1} < u_N$ and $u_{N-1} < u_N$. Unimodality makes this enough, but you then need to say so; it is cleaner to find the sign of every difference.

Beyond STEP

Property L is called *log-concavity*: $\ln u_r$ is a concave function of r. Log-concave sequences turn up all over combinatorics and probability. The binomial coefficients $\binom{n}{r}$ are an example, and Newton's inequalities say that the coefficients of a polynomial with only real roots are log-concave. The recurrence in part (ii) is a linear recurrence with a repeated root, which is why the solution has the form $(A+Br)\alpha^r$, just as $(A+Bx)\mathrm{e}^{\alpha x}$ arises for repeated roots in differential equations. For the differential-equation side of this parallel, see Boyce and DiPrima, *Elementary Differential Equations*; for problems in this spirit, Siklos, *Advanced Problems in Mathematics*.

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